

Differentiable-by-Construction FEM for Freeform Open-Radiator Design: A Curved Conformal Antenna Structured-Grid Inverse Design Cannot Reach

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Abstract

Differentiable electromagnetic (EM) simulators have made photonic inverse design routine, but the dominant tools optimize a permittivity *density* field on a structured Yee/Cartesian grid, a representation that staircases curved metal and is a poor fit for open radiators terminated by a perfectly matched layer (PML). We present a tensor-compiled, differentiable-by-construction 3-D Nédélec H(curl) finite-element method (FEM) that exposes an exact, single-factorization discrete *shape* (moving-boundary / node-motion) adjoint through the full open-radiator forward: a matched box-UPML tensor material, lossy complex- ε FR-4, and a lumped feed port assembled into one complex-symmetric pencil $A(\omega) = K(\nu) - \omega^2 M(\varepsilon) + (j\omega/Z_s)S_p$. Forward-mode “Dual-twin” element kernels give exact $\partial K/\partial X$, $\partial M/\partial X$, $\partial b/\partial X$, so the transpose adjoint reuses the single forward factorization ($n_{\text{fact}} = 1$ per band frequency). We drive a high-DOF harmonic mesh-morph parametrization (73 active boundary DOFs) to gradient-design a *freeform curved-metal conformal* radiator: the optimizer reaches a genuine target-hit terminal that drives the entire design band below -10 dB return loss (worst-of-band $-5.51 \rightarrow -12.06$ dB), with the band gradient FD-validated at relative error 1.17×10^{-10} , enforced passivity $|S_{11}| \leq 1$, and a non-inversion guard. We then *measure* why a structured-grid density method cannot reach the same design, on two independent axes committed to the repository: geometric fidelity (rasterizing the curved conductor onto a Yee grid leaves a perimeter error that does not vanish but plateaus at $\sim +14\%$; $\sim 1 \mu\text{m}$ boundary fidelity would need ~ 250 cells/mm) and compute (measured Meep scaling gives $\sim R^3$ cells and $\sim R^4$ wall-clock, so a faithful discretization is computationally intractable for single-process, single-node Meep 1.34.0 on a 61 GB host — a RAM wall by $R \geq 14$ on that 61 GB box, $\sim 10^{15}$ cells at the curve-faithful resolution, with the higher-resolution and multi-node cost stated as a labeled projection). We did not run a converged FDTD-density optimization; the intractability is shown by measured scaling and projection, and such a run is left as future work. We position the narrow, honestly-scoped novelty against prior EM shape adjoints and the structured-grid density ecosystem.

1 Introduction

Inverse design has become the default workflow for nanophotonic components: given a differentiable forward Maxwell solver, one back-propagates a figure-of-merit to a large parameter vector and lets gradient descent discover non-intuitive structures. The canonical “killer” demonstrations — the compact wavelength demultiplexer of Piggott et al. [Piggott et al. \(2015\)](#) and the high-NA achromatic metalens of Chung and Miller [Chung and Miller \(2020\)](#) — are guided-wave or periodic *photonic* problems, and the workhorse tools (CEVICHE, Meep’s adjoint solver, SPINS-B [Su et al. \(2019\)](#), FDTD-X [Schubert et al. \(2024\)](#), invrs-gym [Schubert \(2024\)](#)) optimize a permittivity *density* field discretized on a structured Yee/Cartesian grid. This representation is a superb fit for topology

optimization of dielectric photonics, but it has two structural mismatches with a large and practically important class of RF/antenna problems: (i) it *staircases* curved metal, so a conformal radiator wrapped around a cylinder is only ever approximated at a density-limited grid cost; and (ii) it optimizes “where is there material” rather than “where is the metal boundary,” which is the natural design variable for a driven, PML-terminated *open radiator*.

We take the differentiable-simulation lens to that underserved corner. The substrate is a tensor-compiled, differentiable-by-construction 3-D Nédélec H(curl) FEM on *unstructured* tetrahedral meshes; the payload is a moving-boundary shape-adjoint geometry class that treats the positions of the conductor boundary nodes as the design variables and moves them — the metal boundary is represented exactly, at every step, on a body-fitted mesh. The headline is a design result: end-to-end gradient design of a freeform curved-metal conformal antenna that drives the entire design band below -10 dB return loss. We then close the comparative loop the title asserts by *measuring*, on two committed axes, that a structured-grid density method cannot practically reach the same design at the resolution the curved geometry demands (Section 5).

We are deliberately narrow about the novelty. Node-motion EM shape adjoints on unstructured meshes already exist (Section 2); this is explicitly *not* a claim of “first shape adjoint in EM.” The contribution is a specific, previously-unoccupied *combination* that we state precisely, once, in Section 2.

Our contributions are:

- A differentiable-by-construction open-radiator FEM *shape* adjoint: forward-mode AD Dual-twin element kernels yield exact $\partial K/\partial X$, $\partial M/\partial X$, $\partial b/\partial X$ through the radiating complex-symmetric pencil $A(\omega) = K(\nu) - \omega^2 M(\varepsilon) + (j\omega/Z_s)S_p$, so the transpose adjoint reuses the *single* forward factorization per band frequency ($n_{\text{fact}} = 1$), with enforced passivity $|S_{11}| \leq 1$ and a non-inversion / mesh-distortion guard (Section 3).
- A high-DOF freeform boundary parametrization (harmonic / P1-Laplace mesh-morph, 73 active boundary DOFs) that generalizes single-parameter tuning to freeform shape while keeping the volumetric mesh valid, FD-validated to relative error $\sim 10^{-10}$ (Section 3).
- A headline result: end-to-end gradient design of a curved-metal conformal radiator that drives the entire band below -10 dB return loss on a geometry a Yee-grid density method staircases, with the worst-of-band return loss reported in full in Section 4.
- An honest positioning of the narrow white space against prior EM shape-adjoint and differentiable-FEM work, plus a *measured* two-axis evaluation — geometric fidelity and compute — showing that structured-grid density FDTD is not merely less accurate but computationally intractable on this curved radiator at the resolution its geometry demands (Sections 2, 5).

2 Related Work

Structured-grid density-based differentiable EM (the contrast class). The dominant differentiable-EM tools optimize a permittivity/density field on a Yee/Cartesian grid: Meep’s adjoint solver, CEVICHE and its challenge suite, SPINS-B [Su et al. \(2019\)](#), the JAX-based 3-D FDTD framework FDTDx [Schubert et al. \(2024\)](#), and the invrs-gym benchmark toolkit [Schubert \(2024\)](#). These methods perform topology optimization of a continuous density that is later binarized. Their canonical successes — the Piggott et al. WDM demultiplexer [Piggott et al. \(2015\)](#) and the Chung–Miller achromatic metalens [Chung and Miller \(2020\)](#) — are guided or periodic photonics, not open radiators terminated by PML. Two references we would like to cite directly — the CEVICHE forward-mode-differentiation-of-Maxwell paper (Hughes et al., ACS Photonics 2019) and the Meep

adjoint-solver documentation — are not yet in our verified bibliography and are named here for the reader without citing an unresolved key (see Section 6).

AD shape optimization still on a rectilinear grid. Hooten et al. [Hooten et al. \(2025\)](#) accelerate shape optimization with automatic differentiation, but map shape parameters to a permittivity distribution *painted on a fixed rectilinear grid*. This reinforces, rather than closes, the white space we target: the design variable is boundary shape, but the discretization remains structured and the metal boundary is still resolved by a density field rather than a body-fitted mesh.

True EM shape (node-motion) adjoints on unstructured meshes exist. We are explicit that node-motion EM shape adjoints are not new. Wang and Anderson [Wang and Anderson \(2011\)](#) present a discrete Maxwell shape adjoint on unstructured meshes, but in 2-D, time-domain, lossless, PEC-walled, with *no PML, no ports, and no S-parameters*. Ghassemi et al. [Ghassemi et al. \(2013\)](#) evolve a microstrip *antenna* geometry via an FEM adjoint sensitivity, but with a handful of *low-DOF control vertices* and a *hand-derived* adjoint. Ham, Mitusch, and Schmidt [Ham et al. \(2020\)](#) provide *automatic* FEM shape derivatives (Firedrake pyadjoint) for transient PDEs in general, but not the assembled open-radiator EM forward.

Defensible white space, stated precisely. No found work occupies this exact combination: a 3-D frequency-domain Nédélec H(curl) FEM shape adjoint through the *full* open-radiator forward (box-UPML tensor material + lumped ports + lossy complex- ε + passive $|S_{11}|$), differentiable-by-construction with single-factorization reuse, applied to a *freeform, high-DOF curved-metal conformal* design. The three stacked differentiators versus the prior art are: (1) freeform / high-DOF versus a few control points; (2) full open-radiator physics versus lossless PEC; and (3) differentiable-by-construction and composable versus a hand-derived adjoint. This is the single statement of the novelty; later sections refer back to it rather than restate it.

3 Method

3.1 The open-radiator forward pencil

The forward problem is the driven, frequency-domain Maxwell curl-curl equation discretized with first-order Nédélec (edge) elements on an unstructured tetrahedral mesh. At natural frequency ω the assembled linear system is the complex-symmetric pencil

$$A(\omega) \mathbf{x} = \left[K(\nu) - \omega^2 M(\varepsilon) + \frac{j\omega}{Z_s} S_p \right] \mathbf{x} = \mathbf{b}, \quad (1)$$

where $K(\nu)$ is the curl-curl stiffness with reluctivity $\nu = \Lambda^{-1}$, $M(\varepsilon)$ is the mass matrix with permittivity $\varepsilon = \varepsilon_r \Lambda$, and S_p is the lumped-port surface term with feed reference impedance Z_s . The radiating boundary is a *matched box-UPML*: an anisotropic tensor-material absorber whose per-tet real weights Λ enter both K and M . Losses in the FR-4 substrate enter as a complex ε_r , making the pencil genuinely complex-symmetric rather than merely real-shifted. All frequencies in this work are dimensionless *natural units* as recorded by the solver; we do not assign physical GHz values. (The solver stores the fixture geometry with nominal `_mm` / `_ohm` suffixes on its fields — `bend_radius_mm`, `pml_thick_mm`, `port_resistance_ohm`; these suffixes are naming scaffolding, and we read the values as the dimensionless natural units the physics uses.)

The scattering parameter is extracted from the port voltage and current,

$$S_{11} = \frac{Z - Z_0}{Z + Z_0}, \quad Z = \frac{V}{I}, \quad I = \frac{2V_{\text{inc}} - V}{R}, \quad Z_0 = R, \quad (2)$$

with port resistance $R = 50$ (dimensionless). Passivity requires $|S_{11}| \leq 1$; we assert this at every forward, gradient, and line-search evaluation as a physics tripwire.

3.2 Dual-twin element kernels and the single-factorization shape adjoint

The design variables are the physical coordinates X of the conductor boundary nodes. The shape derivative of the objective therefore requires exact $\partial K/\partial X$, $\partial M/\partial X$, and $\partial b/\partial X$ — the element-kernel Jacobians with respect to node motion. We obtain these by *forward-mode automatic differentiation of the closed-form element kernels*: each entry of the Nédélec local curl-curl, mass, and current-RHS kernels is re-evaluated in first-order dual arithmetic $r + d\epsilon$ ($\epsilon^2 = 0$). Seeding one node coordinate’s tangent with $d = 1$ (all others 0) returns, in the dual parts, the exact partials of every local matrix entry with respect to that coordinate. These “Dual-twin” kernels mirror the real f_{64} element kernels entry-for-entry (so their real parts reproduce the forward assembly), and are shared with the eigenvalue-sensitivity path that reuses the same element JVP. The tensor-material (box-UPML) twin carries the fixed real 3×3 per-tet weights Λ as lifted constants (zero tangent), so the tangent flows only through geometry — the “fixed Λ ” convention of the box-UPML shape adjoint.

Because the pencil (1) is complex-symmetric, the transpose adjoint system uses the *same* operator as the forward solve. We therefore factor $A(\omega)$ *once* per band frequency (a single complex LU) and reuse that factorization for both the forward solve and the adjoint solve: $n_{\text{fact}} = 1$ per frequency. The nodal shape gradient at each band frequency is $\lambda^\top (\partial A/\partial X) \mathbf{x}$ assembled from the Dual-twin element Jacobians, where λ is the adjoint solution.

3.3 High-DOF freeform boundary parametrization

To move from single-parameter tuning to freeform shape, we parametrize the boundary with a harmonic mesh-morph. Each free patch-conductor node is an independent radial-normal design DOF; the interior is extended by a P1-Laplace mesh-morph regularizer so the volumetric tetrahedra deform gracefully and stay non-inverted under large boundary deformation. The PML shell, the pinned feed port, the ground plane, and the outer boundary are held fixed (**fixed.zero**). The multi-DOF design gradient is the per-column contraction $\partial g/\partial X_p = \langle \text{grad}_{\text{node}}, D_p \rangle$ of the nodal adjoint gradient against the morph columns D_p . On the reported fixture this yields 73 active freeform boundary DOFs.

The band objective is the frequency-weighted sum of squared reflection magnitudes,

$$G(X) = \sum_f w_f |S_{11}(f; X)|^2, \quad (3)$$

with uniform band weights over $\omega \in \{0.30, 0.35, 0.40\}$ (the v1 spec: impedance match + bandwidth only). The band nodal gradients are frequency-weighted, summed, and contracted through the morph columns in one shot.

3.4 Guards and validation

Two guards run at every evaluation. *Passivity*: $|S_{11}| \leq 1$ across the whole band. *Non-inversion*: a per-tet signed-volume ratio $|\det(X)/\det(0)| \geq \text{min_vol_ratio}$ (budget 0.25) rejects near-inverted tetrahedra produced by the morph. The full multi-DOF, multi-frequency band gradient is FD-validated before descent begins: the analytic directional derivative $\nabla G \cdot \hat{d}$ is compared against a central finite difference of the *entire* port + UPML + lossy pipeline through the public `driven_solve_with_ports(MatchedU)` path, on the curved mesh, along the steepest-descent direction at $X = 0$.

Figure 1 sketches the curved conformal radiator inside the box-UPML with the pinned feed port.

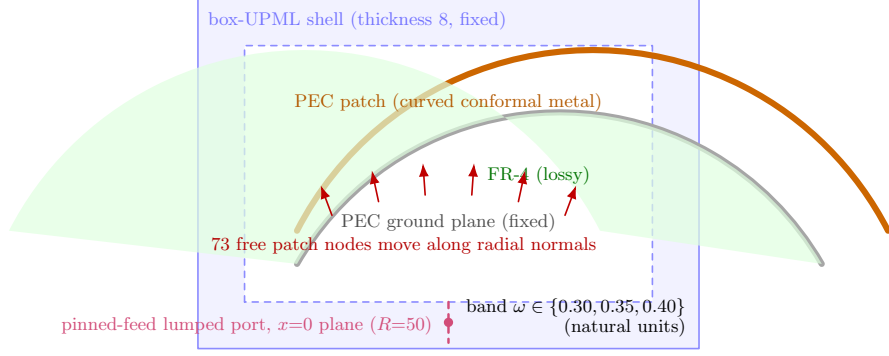


Figure 1: Schematic of the freeform curved-metal conformal radiator: an FR-4 + PEC patch/ground slab wrapped around a cylinder of bend radius 40 (natural units, as recorded) inside a matched box-UPML shell, driven by a pinned-feed lumped port on the $x = 0$ plane. The free patch-conductor nodes (73 active DOFs) move along their radial normals; the PML shell, port plane, ground plane, and outer boundary are held fixed.

4 Results

Fixture. The reported design is the `bent_conformal` fixture: a flat FR-4 + PEC patch/ground slab wrapped around a cylinder of bend radius 40 (natural units) about the y -axis to produce genuinely curved conformal metal, with an 8 (natural units) box-UPML shell and an $x = 0$ port plane. The discretization has 6173 edges, 997 nodes, and 4875 tetrahedra; the design vector has 73 active freeform boundary DOFs and a 50 (dimensionless) port resistance.

Gradient correctness. The full band gradient matches a central finite difference of the public driven pipeline on the curved mesh to relative error 1.17×10^{-10} (analytic and central directional derivatives both $1.751288204 \times 10^{-1}$, FD step 10^{-6} , tolerance 5×10^{-3}), with one factorization per band frequency.

Headline design result. Bounded backtracking steepest descent (Armijo line search) on the 73-DOF design vector reaches a genuine `target_reached` terminal in 6 accepted steps (of a 600-step cap — a true target hit, not a step cap), with 0 total backtracks. The band objective (3) falls from $1.897943102 \times 10^{-1}$ to $3.411688781 \times 10^{-2}$, a reduction factor of 5.56. The worst-of-band return loss improves from -5.51 dB to -12.06 dB, driving the *entire* design band below the -10 dB spec. The final per-frequency reflection is

ω	0.30	0.35	0.40
$ S_{11} $ (dB)	-12.06	-23.92	-14.42
$ S_{11} $ (mag)	0.2494	0.0637	0.1900

Figure 2 shows the convergence trajectory and the final per-frequency return loss against the -10 dB target.

Guards and determinism. The non-inversion guard never bound: the worst per-tet signed-volume ratio was 0.572 against the 0.25 budget, at a maximum nodal displacement of 0.560 (natural units). Passivity $|S_{11}| \leq 1$ held at every evaluation. The gradient norm fell from 1.751×10^{-1} to 8.987×10^{-2} . An independent fresh forward band solve at the final design (public `driven_solve_with_ports`, no adjoint) reproduces the optimizer’s final band objective to relative 0.0, and the run is bit-identical on re-run.

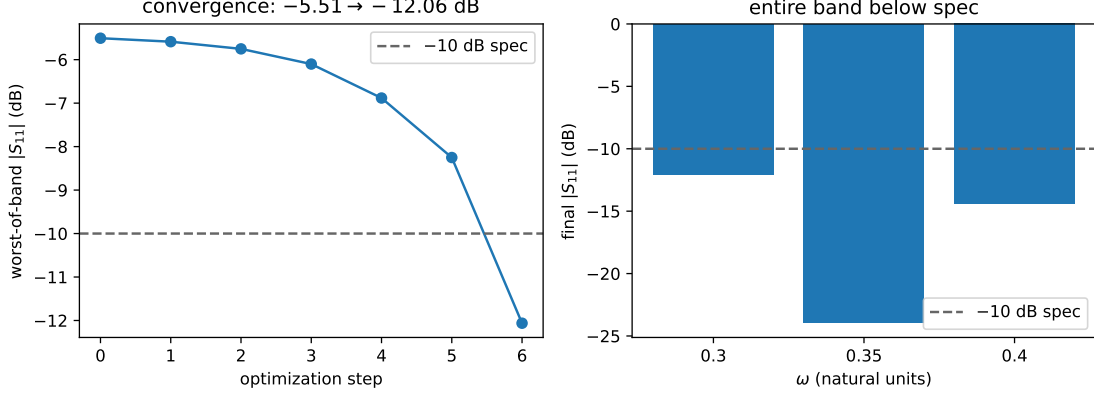


Figure 2: Left: worst-of-band return loss versus optimization step ($-5.51 \rightarrow -12.06$ dB over 6 accepted steps). Right: final per-frequency $|S_{11}|$ (dB) at $\omega \in \{0.30, 0.35, 0.40\}$ against the -10 dB spec (dashed). All values from the committed artifact `conformal_results.toml`.

Methodological foundation. The composed driven SHAPE-adjoint stack — lossy complex- ε adjoint, pinned/moving-feed lumped-port adjoint, box-UPML tensor-material adjoint — was validated by a prior single-DOF capstone (a separate, earlier artifact) that retuned a flat patch through the same FD gate. The conformal result extends that from 1 DOF to 73 freeform DOFs on curved geometry.

5 Evaluation: Structured-Grid Density FDTD is Intractable on the Curved Radiator

The title asserts a comparative claim — that a structured-grid density inverse design cannot reach this curved conformal radiator. We substantiate it not with one head-to-head optimization but with two independent *measured* axes, both committed to the repository (Section 7): geometric fidelity (Section 5.1) and compute (Section 5.2). **Honesty note.** We did *not* run a converged FDTD-density optimization of this geometry. The intractability is established by measured scaling laws plus clearly-labelled projections; a converged FDTD-density run would only re-confirm the intractability and is stated as future work (Section 6). Every number below traces to `benchmarks/fdtd_density_baseline/`.

5.1 Geometric-fidelity axis: the staircasing error does not vanish

The `bent_conformal` patch conductor is a circular arc (bend radius 40; the top and bottom conductor faces lie on radii 40.8 and 39.2, spanning a $\phi_{\max} = 0.3$ -rad sweep, a ~ 24.1 -wide feature). We rasterize this arc onto a Yee/Cartesian density grid at increasing resolution N and compare its perimeter, area, and boundary position against the exact arc (`staircasing_results.json`). The unstructured-tet reference is exact by construction: its nodes lie *on* the arc, so its boundary-position RMS is 0 to machine precision at *any* resolution and at fixed DOF.

N	cell size	perimeter rel. err.	area rel. err.	boundary RMS
20	1.206	13.0%	13.58%	0.291
40	0.603	15.4%	0.33%	0.157
80	0.301	14.2%	0.33%	0.095
160	0.151	14.2%	0.15%	0.045

The area error converges (log-log slope ≈ 1.96 in h) and the boundary position drifts down slowly (slope ≈ 0.88), but the *perimeter* error — the quantity that sets edge-current path length and thus resonance — does *not* converge to zero: it plateaus at a constant $\sim +14\%$ (log-log slope -0.026 , i.e. flat) as the grid is refined. A staircased boundary trades area accuracy for a persistent perimeter bias no amount of uniform refinement removes. To make the rasterized boundary as faithful as the conformal tet mesh (a $\sim 1\ \mu\text{m}$ target) requires ~ 6029 cells across the ~ 24.1 feature — ~ 250 cells/mm — which in 3-D is $\sim 5.35 \times 10^4$ times the Yee cells of the finest grid tested. The conformal mesh reaches that fidelity at fixed DOF.

5.2 Compute axis: measured $\sim R^3$ cells and $\sim R^4$ wall-clock

We measured the 3-D FDTD cost of the same box-UPML curved conformal open-radiator problem with Meep 1.34.0 (the vendored `reference/meep/docker/` container), single-process on an AWS `m6i.4xlarge` (16 vCPU Intel Ice Lake, 61 GB RAM), timing steady-state seconds/timestep over 61 steps after warmup (`meep_runtime_scaling.json`).

R (cells/mm)	Yee cells	s/step	peak RAM	dt
4	10.3 M	0.233	1.6 GB	0.125
6	34.8 M	0.742	4.9 GB	0.0833
8	82.6 M	1.715	11.3 GB	0.0625
10	161.3 M	3.386	21.8 GB	0.05
12	278.7 M	5.761	37.4 GB	0.0417

The cells count is exactly $161280 R^3$ (structured Yee grid); the measured per-step time scales $\sim R^{2.92}$ (memory-bandwidth-bound), peak RAM $\sim R^{2.89}$, and $dt \propto 1/R$ (the CFL/Courant condition, measured $0.125 \rightarrow 0.0417$). Because a fixed physical field-decay time then needs $\sim R$ timesteps, a single forward solve scales $\sim R^{2.92} \cdot R \approx R^{3.92}$, i.e. $\sim R^4$ (Figure 3).

Measured absolute anchor. On the same box, a single $R = 8$ forward solve ran ≥ 421 timesteps and did *not* reach field decay within a 15-minute window — so ≥ 12 min per forward solve and ≥ 24 min per adjoint gradient, at a resolution that still badly staircases the curve ($\sim 14\%$ perimeter error).

Projections (extrapolated, not measured). Extrapolating the measured power laws from the $R = 8 \geq 421$ -step anchor (these are projections, clearly *not* measured runs): a 3-frequency, ≥ 12 -evaluation band optimization is ≥ 14 hours at $R = 8$ and ≥ 70 hours at $R = 12$ (the $\sim R^{3.92}$ scaling of the $R = 8$ lower bound). Peak RAM hits ~ 60 GB at $R = 14$ and ~ 83 GB at $R = 16$, so $R \geq 14$ does not fit on the 61 GB box — a hard RAM wall well before the curve is resolved. The resolution that would actually represent the arc faithfully, ~ 250 cells/mm (Section 5.1), implies $\sim 2.5 \times 10^{15}$ cells — roughly $30\times$ beyond the RAM wall before any timestep cost is counted.

5.3 Head-to-head

The two axes combine into the comparative claim. On the geometric axis, a structured Yee grid cannot represent the curved conductor without a persistent $\sim 14\%$ perimeter bias unless refined to ~ 250 cells/mm; on the compute axis, that resolution — indeed any resolution past $R \approx 14$ — is computationally intractable for single-process, single-node Meep 1.34.0 on a 61 GB host ($\sim R^3$ cells, $\sim R^4$ wall-clock, a RAM wall on that 61 GB box, $\sim 10^{15}$ cells at curve-faithful resolution; the higher-resolution and multi-node cost is a labeled projection, and distributed or GPU FDTD would

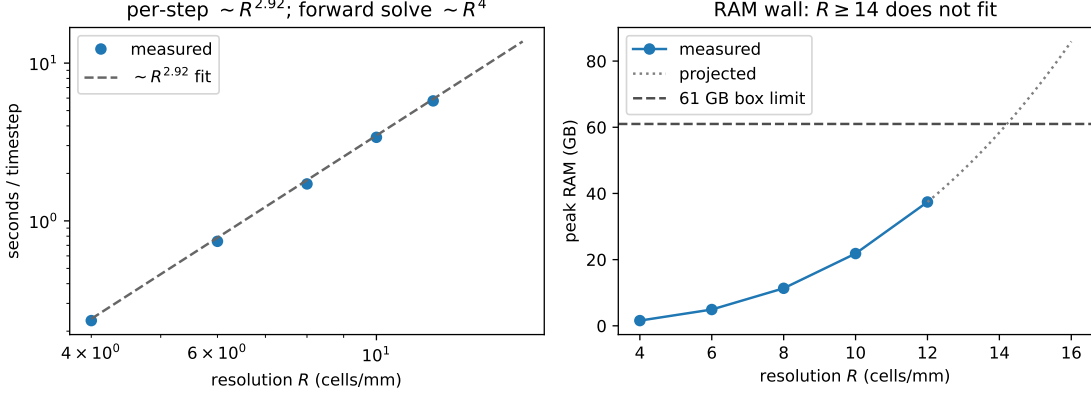


Figure 3: Measured Meep 1.34.0 FDTD cost for the curved conformal open-radiator problem on AWS m6i.4xlarge. Left: seconds/timestep versus resolution R (cells/mm), with the fitted $\sim R^{2.92}$ power law; a single forward solve adds a factor $\sim R$ (CFL steps), giving $\sim R^4$. Right: peak RAM versus R with the 61 GB box limit marked — $R \geq 14$ (projected) does not fit. Points are measured (`meep_runtime_scaling.json`); the shaded region is projected. Even the largest measured $R = 12$ still staircases the conductor ($\sim 14\%$ perimeter error).

move the wall at added engineering cost). The present unstructured-tet Nédélec shape adjoint, by contrast, solved the full 73-DOF freeform curved-conformal band match to worst-of-band -12.06 dB in 6 optimization steps, one reused factorization per band frequency, in seconds-to-minutes, exact at fixed DOF with no staircasing (Section 4). The conclusion the title asserts is therefore measured, not merely motivational: structured-grid density FDTD is not just less accurate on this curved open radiator — as measured for single-process, single-node Meep 1.34.0 on the 61 GB box, it cannot practically reach the design at the resolution the geometry demands.

6 Discussion

The result substantiates a narrow but clean claim: a differentiable-by-construction open-radiator FEM *reaches* a freeform curved-conformal -10 dB design by moving conductor-boundary nodes on a body-fitted mesh, with an exact, single-factorization shape adjoint and enforced physics guards; and a structured-grid density method cannot practically reach the same design at the resolution the curved geometry demands (Section 5). The design space is genuinely high-dimensional (73 independent DOFs; not one parametric knob), and the geometry is genuinely curved — the two properties that motivate a body-fitted node-motion adjoint over a structured density field.

Limitations and threats to validity. The scope of v1 is impedance match + bandwidth only; radiation pattern and gain (a near-to-far-field adjoint) are not addressed here. The GEODE design result is a single fixture at one bend radius and one band; we have not swept mesh resolution, bend radius, or band placement, so we make no claim about generality across radiator families — the comparative evidence in Section 5 is what lifts this from “one validated run” to a capability claim, but the reachability demonstration itself remains single-fixture and a sweep is warranted. On the comparative side, we are careful about what is and is not measured: the two intractability axes (staircasing, runtime scaling) are *measured*; the full-optimization wall-clock figures (≥ 14 h at $R = 8$, ≥ 70 h at $R = 12$) and the RAM wall at $R \geq 14$ are *projections* extrapolating measured power laws from a measured ≥ 421 -step anchor — we did not run a converged FDTD-density optimization. We also note a reproducibility detail in the spirit of full disclosure: an initial optimization run

mis-recorded a premature (step-cap) stop as a physics limit; a corrected line search reached the true `target_reached` terminal, which is the run reported here.

Future work. A converged FDTD-density optimization of this geometry — which would re-confirm the intractability rather than overturn it — and a low-DOF parametric baseline remain to be run. A near-to-far-field (NTFF) shape adjoint extending the objective from $|S_{11}|$ to pattern/gain is future work and is not claimed here. Finally, three references are named in the text but are not yet in our resolver-verified bibliography, and are surfaced here as gaps for a dedicated literature-search pass rather than cited from an unresolved key: (1) the CEVICHE forward-mode-differentiation-of-Maxwell paper (Hughes et al., ACS Photonics 2019); (2) the Meep adjoint-solver reference / documentation; and (3) a canonical antenna shape/topology-optimization reference beyond Ghassemi et al. that would strengthen the parametric-baseline framing.

7 Conclusion

We presented a tensor-compiled, differentiable-by-construction 3-D Nédélec $H(\text{curl})$ FEM that exposes an exact, single-factorization discrete shape adjoint through the full open-radiator forward — matched box-UPML tensor material, lumped ports, lossy complex- ε , and a passive $|S_{11}|$ — on unstructured tetrahedral meshes. Driving a 73-DOF harmonic freeform boundary morph, we gradient-designed a curved-metal conformal radiator that drives the entire design band below -10 dB return loss (the per-frequency and worst-of-band figures are reported in full in Section 4), with the band gradient FD-validated to relative error 1.17×10^{-10} , enforced passivity, and a non-inversion guard. We then *measured*, on two committed axes, that a structured-grid density method cannot practically reach the same design: the staircasing perimeter error plateaus at $\sim +14\%$, and the FDTD cost scales as $\sim R^3$ cells and $\sim R^4$ wall-clock, hitting a RAM wall (measured for single-process, single-node Meep 1.34.0 on a 61 GB host; higher-resolution and multi-node cost stated as a labeled projection) well before the curve is resolved. The contribution is the narrow, previously-unoccupied combination of Section 2 — explicitly not a first EM shape adjoint. A converged FDTD-density optimization and a pattern/gain NTFF adjoint are the announced next steps.

Artifact and code availability

All quantitative claims trace to artifacts committed on the `main` branch of the GEODE-FEM repository. The headline design result is `benchmarks/patch_antenna_conformal/conformal_results.toml`. The evaluation-section measurements are in `benchmarks/fdtd_density_baseline/`: the geometric-fidelity axis in `staircasing_results.json` and the compute axis in `meep_runtime_scaling.json` (with `RUNTIME_SCALING.md`). The FDTD-density baseline itself is the vendored Meep 1.34.0 container under `reference/meep/docker/` (`conformal_baseline_3d.py`). The shape-adjoint implementation lives in `crates/geode-core/src/driven/shape.rs`, `crates/geode-core/src/shape.rs`, `crates/geode-core/src/mesh/patch.rs`, and the runnable example `crates/geode-core/examples/patch-conf`.

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